

Chapter 12

The Einstein Correspondence Principle

Recovery of General Relativity from the Theory of Entropicity

12.1 Introduction

Every successful physical theory must reproduce the verified predictions of earlier theories within their domains of validity.

Special Relativity reproduces Newtonian mechanics when

$$v \ll c.$$

Quantum mechanics reproduces classical mechanics when

$$\hbar \rightarrow 0.$$

Likewise, the Theory of Entropicity must reproduce General Relativity under appropriate physical conditions.

This requirement is not merely desirable.

It is essential.

Without such correspondence, the Theory of Entropicity could not claim consistency with the enormous body of experimental evidence supporting Einsteinian gravitation.

The purpose of this chapter is to establish this correspondence rigorously.

We shall demonstrate that General Relativity emerges naturally as a limiting case of the Theory of Entropicity when entropy gradients become negligible and informational sectors cease contributing dynamically.

This result is referred to as the **Einstein Correspondence Principle**.

12.2 The Principle of Correspondence

The Einstein Correspondence Principle may be stated as follows.

Einstein Correspondence Principle

In the near-equilibrium limit of the Theory of Entropicity, the Obidi Field Equations reduce to the Einstein Field Equations of General Relativity.

Symbolically,

$$\text{ToE} \longrightarrow \text{GR}$$

under appropriate limiting conditions.

This principle serves as one of the primary consistency requirements of the theory.

12.3 The Obidi Field Equations

The master field equations of ToE are

$$G_{(\mu\nu)J} + \Lambda_{(J)}g_{(\mu\nu)J} = 8\pi G_{(J)}T_{(\mu\nu)J}^{ent}.$$

Applying the Obidi Addition Rule yields

$$\sum_J (G_{(\mu\nu)J} + \Lambda_{(J)}g_{(\mu\nu)J}) = 8\pi \sum_J G_{(J)}T_{(\mu\nu)J}^{ent}.$$

Since

$$J \in \{FR, FS, L\},$$

the complete dynamics contains contributions from:

- Fisher–Rao geometry,
- Fubini–Study geometry,
- Lorentzian geometry.

General Relativity corresponds to retaining only the Lorentzian contribution.

12.4 The Near-Equilibrium Limit

The essential assumption is that the entropy field approaches equilibrium.

Mathematically,

$$\nabla_\mu S \rightarrow 0.$$

This implies

$$(\nabla S)^2 \rightarrow 0.$$

and

$$\nabla_\mu \nabla_\nu S \rightarrow 0.$$

The entropy field becomes approximately constant throughout the manifold.

Denoting the equilibrium value by

$$S_0,$$

one obtains

$$S(x) \rightarrow S_0.$$

The entropy field therefore loses its dynamical influence.

12.5 Consequences for the Local Obidi Action

The Local Obidi Action is

$$I_{LOA} = \sum_J \int \left(R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} (\nabla S)^2 + V(S) \right) e^{S/k_B} \sqrt{|g^{(J)}|} d^4 x.$$

At equilibrium,

$$(\nabla S)^2 = 0.$$

Consequently the kinetic contribution vanishes.

Furthermore,

$$e^{S/k_B} = e^{S_0/k_B} = \text{constant}.$$

The exponential factor may therefore be absorbed into an overall normalization constant.

The action becomes

$$I_{LOA} \rightarrow e^{S_0/k_B} \sum_J \int \left(R^{(J)} - \Lambda_{(J)} + V(S_0) \right) \sqrt{|g^{(J)}|} d^4 x.$$

The explicit dynamical role of entropy disappears.

12.6 Decoupling of Informational Sectors

The Theory of Entropicity assumes that the Fisher–Rao and Fubini–Study sectors are activated by informational and entropic structure.

At equilibrium,

$$\nabla S = 0,$$

the informational curvature contributions become negligible.

Thus

$$G_{\mu\nu}^{(FR)} \rightarrow 0,$$

and

$$G_{\mu\nu}^{(FS)} \rightarrow 0.$$

Only the Lorentzian sector remains dynamically relevant.

This yields

$$G_{(\mu\nu)L} + \Lambda_L g_{(\mu\nu)L} = 8\pi G_L T_{(\mu\nu)L}.$$

12.7 Emergence of the Lorentzian Sector

The Lorentzian sector becomes the effective geometry of the system.

We identify

$$g_{(\mu\nu)L} = g_{\mu\nu}.$$

Similarly,

$$G_{(\mu\nu)L} = G_{\mu\nu}.$$

The Lorentzian sector therefore assumes the role ordinarily occupied by spacetime in General Relativity.

The informational sectors become hidden degrees of freedom.

12.8 Reduction of the Obidi Action

Under these assumptions the Local Obidi Action becomes

$$I_{LOA} \rightarrow C \int (R - 2\Lambda) \sqrt{|g|} d^4x,$$

where

$$C = e^{S_0/k_B}$$

is a constant normalization factor.

Since overall multiplicative constants do not affect the variational equations,

$$C$$

may be absorbed into the definition of the action.

Consequently,

$$I_{LOA} \rightarrow I_{EH}.$$

The Einstein–Hilbert Action is recovered.

12.9 Recovery of Einstein's Equations

Variation of the reduced action yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

This is precisely Einstein's Field Equation.

No additional assumptions are required.

The recovery follows directly from the equilibrium limit of the Theory of Entropicity.

12.10 The Entropic Origin of Einstein Gravity

The preceding derivation admits a profound interpretation.

General Relativity is not fundamental.

It is emergent.

The Einstein equations arise when:

1. Entropy becomes approximately uniform.
2. Information gradients disappear.
3. Informational sectors decouple.
4. Only spacetime geometry remains visible.

Thus Einstein gravity represents the macroscopic equilibrium state of a deeper entropic theory.

12.11 The Einstein–Obidi Correspondence Map

The relationship between ToE and General Relativity may be summarized through the following correspondences.

Hybrid Manifold

$$HMAS \rightarrow M.$$

The hybrid manifold reduces to an ordinary spacetime manifold.

Sectoral Metrics

$$g_{\mu\nu}^{(FR)} + g_{\mu\nu}^{(FS)} + g_{\mu\nu}^{(L)} \rightarrow g_{\mu\nu}.$$

Only the Lorentzian metric survives.

Obidi Einstein Tensor

$$G_{(\mu\nu)J} \rightarrow G_{\mu\nu}.$$

Entropic Stress-Energy

$$T_{\mu\nu}^{ent} \rightarrow T_{\mu\nu}.$$

Local Obidi Action

$$I_{LOA} \rightarrow I_{EH}.$$

Obidi Field Equations

$$G_{(\mu\nu)J} + \Lambda_{(J)}g_{(\mu\nu)J} = 8\pi G_{(J)}T_{(\mu\nu)J}^{ent}$$

becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

12.12 Physical Interpretation

The Einstein Correspondence Principle implies that General Relativity occupies a role analogous to thermodynamics.

Thermodynamics describes macroscopic equilibrium behavior.

Statistical mechanics describes microscopic dynamics.

Similarly,

General Relativity

describes equilibrium gravitational behavior,

while

Theory of Entropicity

describes the deeper entropic dynamics from which gravity emerges.

This analogy is one of the most important conceptual consequences of the theory.

12.13 Recovering Classical Tests of General Relativity

Because the Einstein equations emerge in the equilibrium limit, all standard predictions of General Relativity are automatically recovered.

These include:

Mercury Perihelion Precession

$$\Delta\phi = 43'' \text{ per century.}$$

Gravitational Light Deflection

$$\delta = 1.75''.$$

Gravitational Redshift

$$\frac{\Delta\nu}{\nu} = \frac{\Delta\Phi}{c^2}.$$

Shapiro Time Delay

$$\Delta t = \frac{2GM}{c^3} \ln \left(\frac{4r_1 r_2}{b^2} \right).$$

Gravitational Waves

Solutions of

$$\square h_{\mu\nu} = 0.$$

Since General Relativity emerges from ToE, these results must also emerge in the appropriate limit.

12.14 The Meaning of Emergence

The word "emergence" is often used loosely.

Within ToE it possesses a precise meaning.

A structure is emergent when:

1. It appears at large scales.
2. It disappears at deeper levels of description.

3. Its governing equations arise from more fundamental equations.

General Relativity satisfies all three criteria within the framework of ToE.

12.15 Conceptual Significance

The Einstein Correspondence Principle accomplishes three objectives.

First, it establishes mathematical consistency with General Relativity.

Second, it explains why Einstein's equations are so successful experimentally.

Third, it provides an interpretation of General Relativity as an equilibrium manifestation of deeper entropic dynamics.

The principle therefore occupies a central position within the Theory of Entropicity.

12.16 Summary

This chapter has established the Einstein Correspondence Principle.

The principal conclusions are:

1. The Local Obidi Action reduces to the Einstein–Hilbert Action.
2. Informational sectors decouple in the equilibrium limit.
3. The Lorentzian sector becomes dominant.
4. The Obidi Field Equations reduce to Einstein's Field Equations.
5. General Relativity emerges as the equilibrium approximation of the Theory of Entropicity.

Symbolically,

Entropy Equilibrium $\longrightarrow$ General Relativity.

This correspondence demonstrates that ToE contains Einstein gravity as a natural limiting case.

12.17 Transition to Cosmology

Having established the recovery of General Relativity, the next question concerns the large-scale structure of the universe.

If entropy is the fundamental organizing principle of nature, what role does it play in cosmology?

Can cosmic expansion, dark energy, and large-scale structure emerge from entropic dynamics?

The next chapter develops the cosmological implications of the Theory of Entropicity and examines the Entropic Cosmological Principle governing the evolution of the universe.